MA 550: Homework 1

Axel Fraud

Problem 1

Consider the two-dimensional Stokes equations

$$\mu\nabla^2\mathbf{u} - \nabla P = \mathbf{b}, \qquad \nabla \cdot \mathbf{u} = 0,$$

where

$$\mathbf{u} = \begin{bmatrix} U \\ V \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} f \\ g \end{bmatrix}, \qquad \mu = 1.$$

The manufactured analytical solutions are

$$U_{\text{an}}(x, y) = \sin(2\pi y)\sin(2\pi x),$$

$$V_{\text{an}}(x, y) = -3.5 + \cos(2\pi x)\left[\cos(2\pi y) - 1\right],$$

and

$$P_{\text{an}}(x, y) = \sin(2\pi y)\cos(2\pi x).$$

The velocity and pressure are periodic on the left and right boundaries. On the top and bottom boundaries,

$$U = 0, \qquad V = -3.5.$$

Solution

The Stokes equations were discretized on a MAC staggered grid. Pressure unknowns were placed at cell centers, the horizontal velocity U was placed on vertical cell faces, and the vertical velocity V was placed on horizontal cell faces. This staggering allows the pressure gradients and velocity divergence to be approximated naturally at their corresponding grid locations.

Using second-order centered finite differences, the discrete Stokes system was assembled in block form as

$$\begin{bmatrix} \mu L_U & 0 & -G_x \\ 0 & \mu L_V & -G_y \\ D_x & D_y & 0 \end{bmatrix} \begin{bmatrix} U \\ V \\ P \end{bmatrix} = \begin{bmatrix} f_{\text{bc}} \\ g_{\text{bc}} \\ h_{\text{bc}} \end{bmatrix}.$$

Here, L_U and L_V are the discrete Laplacian operators, G_x and G_y are the discrete pressure-gradient operators, and D_x and D_y form the discrete divergence operator. Periodicity in the x-direction was implemented directly through periodic indexing. The prescribed

velocity values on the top and bottom boundaries were incorporated into the appropriate right-hand-side terms.

Since pressure in the Stokes equations is determined only up to an additive constant, the discrete system contains a pressure null space. To obtain a unique numerical solution, one pressure degree of freedom was fixed while solving the linear system. After solving, the mean numerical pressure was subtracted before comparison with the manufactured pressure:

$$P_{\text{num}} \leftarrow P_{\text{num}} - \frac{1}{N_P} \sum_{i=1}^{N_P} P_{\text{num},i}.$$

As a consistency check, the discrete gradient and divergence operators satisfied

$$\left\| G_x + D_x^T \right\|_{\max} = 0, \qquad \left\| G_y + D_y^T \right\|_{\max} = 0.$$

This verifies the expected discrete adjoint relationship between the gradient and divergence operators.

The computed velocity components had the ranges

$$-0.998970 \le U \le 0.998970,$$

and

$$-5.497939 \le V \le -1.502061.$$

The corresponding velocity magnitude satisfied

$$1.508223 \le \|\mathbf{u}\| \le 5.491793.$$

Because U and V are stored at different locations on the staggered grid, the two components were interpolated to the pressure-cell centers before computing the velocity magnitude and plotting the streamlines. The resulting numerical velocity field is shown in Figure 1.

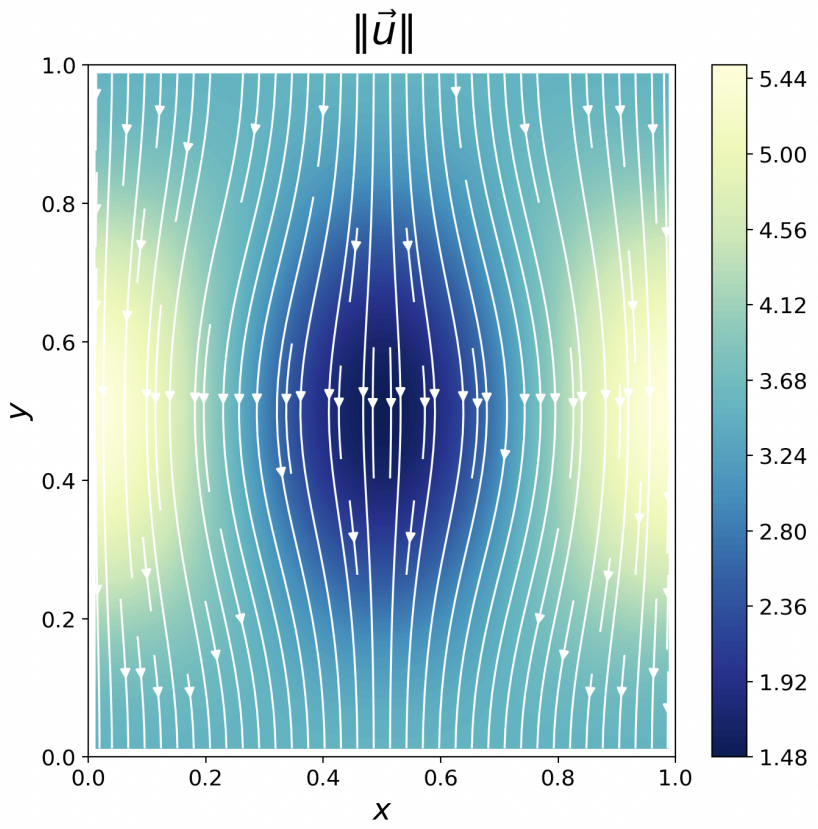


Figure 1: Numerical velocity magnitude and streamlines for the manufactured Stokes solution.

The numerical solution agrees closely with the manufactured solution. The maximum absolute errors were

$$\max |U_{\text{num}} - U_{\text{an}}| = 1.023963 \times 10^{-3},$$
$$\max |V_{\text{num}} - V_{\text{an}}| = 3.076324 \times 10^{-3},$$

and

$$\max |P_{\text{num}} - P_{\text{an}}| = 1.022491 \times 10^{-3}.$$

The streamline pattern in Figure 1 is consistent with the manufactured velocity field. The small errors in all three solution variables, together with the operator consistency checks, indicate that the staggered-grid Stokes discretization is correctly reproducing the analytical solution.

Problem 2

Perform an error analysis by solving the Stokes system on increasingly fine grids with

$$N_x = N_y.$$

For each grid, compute the relative L^2 error. For example, the error in U is defined by

$$\mathrm{err}_U = \frac{\|U_{\mathrm{num}} - U_{\mathrm{an}}\|_2}{\|U_{\mathrm{an}}\|_2}.$$

The same relative error was computed for V and P. The goal is to demonstrate second-order convergence as the grid is refined.

Solution

The Stokes solver from Problem 1 was applied on a sequence of uniform grids with

$$N_x = N_y = N,$$

where

$$N = 10, 20, 30, \ldots, 100.$$

For each grid resolution, the numerical velocity components were first interpolated to the pressure-cell centers so that the numerical and manufactured solutions were evaluated at the same spatial locations. As in Problem 1, the mean of the numerical pressure was removed before computing the pressure error.

The relative L^2 errors were calculated using

$$E_U = \frac{\|U_{\mathrm{num}} - U_{\mathrm{an}}\|_2}{\|U_{\mathrm{an}}\|_2}, \qquad E_V = \frac{\|V_{\mathrm{num}} - V_{\mathrm{an}}\|_2}{\|V_{\mathrm{an}}\|_2},$$

and

$$E_P = \frac{\|P_{\mathrm{num}} - P_{\mathrm{an}}\|_2}{\|P_{\mathrm{an}}\|_2}.$$

To measure the observed convergence rate between consecutive grids, the order was calculated from

$$p = \frac{\log(E_{N_1}/E_{N_2})}{\log(h_{N_1}/h_{N_2})}, \qquad h_N = \frac{1}{N}.$$

For a second-order discretization, we expect

$$E = \mathcal{O}(h^2),$$

and therefore an observed convergence order approaching

$$p = 2.$$

First- and second-order reference curves were also included in the log–log error plot. The reference curves were scaled from the initial U error according to

$$E_{\mathrm{first}} = E_U(10)\left(\frac{10}{N}\right),$$

and

$$E_{\mathrm{second}} = E_U(10)\left(\frac{10}{N}\right)^2.$$

The resulting convergence behavior is shown in Figure 2.

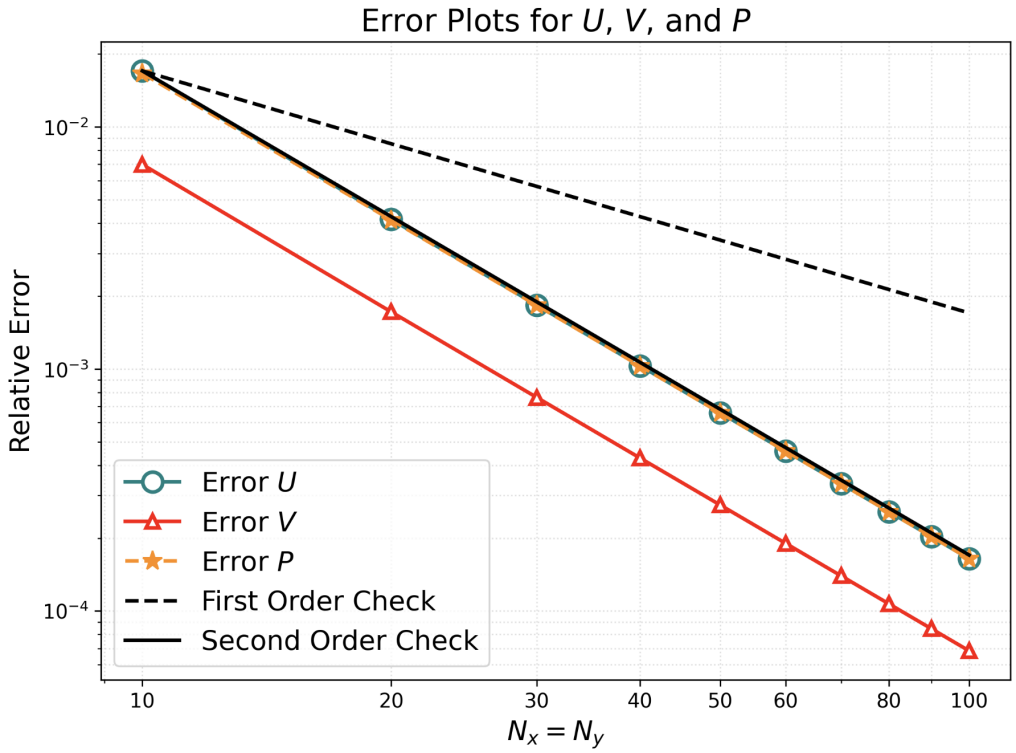


Figure 2: Relative L^2 errors for U, V, and P under grid refinement, together with first- and second-order reference curves.

As the grid is refined, the errors in all three solution variables decrease consistently. The U and P errors are nearly identical over the tested grid resolutions, which causes their curves to nearly overlap in Figure 2. The V error is smaller in magnitude, but follows the same convergence trend.

Most importantly, the numerical error curves are approximately parallel to the second-order reference curve and decrease substantially faster than the first-order reference curve. The observed orders provide a quantitative confirmation of this behavior. For the initial refinement from $N = 10$ to $N = 20$, the observed orders were

$$p_U = 2.0374, \qquad p_V = 2.0233, \qquad p_P = 2.0125.$$

As the grid was refined, these values approached 2. For the final refinement from $N = 90$ to $N = 100$, the observed orders were

$$p_U = 2.0008, \qquad p_V = 2.0005, \qquad p_P = 2.0003.$$

Therefore,

$$E_U = \mathcal{O}(h^2), \qquad E_V = \mathcal{O}(h^2), \qquad E_P = \mathcal{O}(h^2).$$

This demonstrates the expected second-order convergence of the staggered-grid finite-difference discretization for the manufactured Stokes problem.

Problem 3: Charlie's Rule

For this problem, I chose to explore making the Stokes solver more flexible by applying it to a different set of boundary conditions.

Lid-Driven Stokes Cavity

Rather than using the periodic boundary conditions and manufactured solution from Problems 1 and 2, I modified the solver to model a lid-driven cavity. The computational domain remained the unit square,

$$\Omega = [0,1] \times [0,1],$$

and the governing equations remained

$$\mu \nabla^2 \mathbf{u} - \nabla P = \mathbf{0}, \qquad \nabla \cdot \mathbf{u} = 0,$$

with

$$\mu = 1.$$

The upper boundary of the cavity was prescribed to move horizontally with unit velocity,

$$U(x,1) = 1, \qquad V(x,1) = 0,$$

while the remaining three walls were stationary with no-slip boundary conditions,

$$U = V = 0.$$

This creates a substantially different problem from the manufactured solution used previously. In Problems 1 and 2, the left and right boundaries were periodic and the forcing terms were constructed from a known analytical solution. Here, there is no manufactured velocity field driving the solution. Instead, the motion of the upper wall drives the flow inside the cavity.

Numerical Implementation

I retained the MAC staggered-grid structure from the original Stokes solver, with pressure at cell centers and the two velocity components located on their corresponding cell faces. The main modification was the treatment of the boundaries.

The periodic indexing used previously in the horizontal direction was replaced by wall boundary conditions. The stationary left, right, and bottom walls enforce zero velocity, while the upper wall imposes the nonzero horizontal lid velocity. These prescribed values were incorporated into the finite-difference equations through the boundary contributions to the right-hand side.

The resulting discrete system retained the same general Stokes block structure,

$$\begin{bmatrix} \mu L_U & 0 & -G_x \\ 0 & \mu L_V & -G_y \\ D_x & D_y & 0 \end{bmatrix} \begin{bmatrix} U \\ V \\ P \end{bmatrix} = \begin{bmatrix} f_{\text{bc}} \\ g_{\text{bc}} \\ h_{\text{bc}} \end{bmatrix},$$

but the operators and boundary contributions now correspond to the nonperiodic cavity walls. As before, the pressure is determined only up to an additive constant, so a pressure degree of freedom was fixed to remove the null space.

After solving the system, the staggered velocity components were interpolated to the pressure-cell centers. The velocity magnitude was then computed as

$$\|\mathbf{u}\| = \sqrt{U^2 + V^2}.$$

Figure 3 shows the resulting velocity magnitude and streamlines.

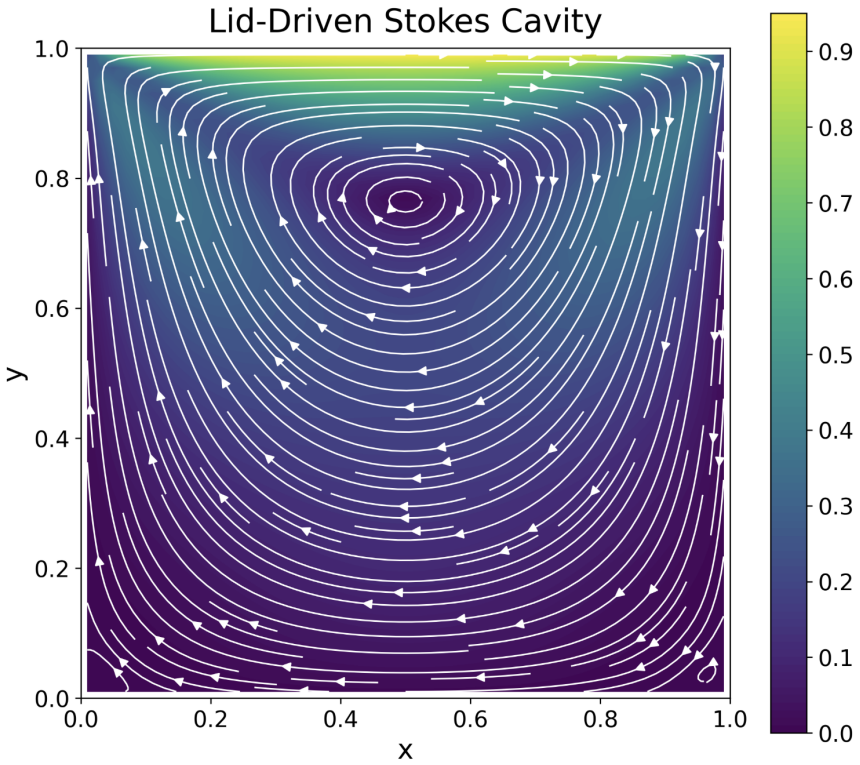


Figure 3: Velocity magnitude and streamlines for the lid-driven Stokes cavity.

The effect of the moving upper wall is clearly visible. The largest velocity magnitudes occur near the lid, where the prescribed horizontal velocity directly drives the fluid. The motion of the lid produces a large clockwise recirculation within the cavity. Moving away from the lid, the velocity decreases as the flow approaches the stationary no-slip walls.

Vorticity

To further explore the numerical solution, I also computed the two-dimensional vorticity field. For a velocity field

$$\mathbf{u} = (U, V),$$

the scalar vorticity is

$$\omega = \frac{\partial V}{\partial x} - \frac{\partial U}{\partial y}.$$

The derivatives were approximated numerically from the cell-centered velocity components. The resulting vorticity field is shown in Figure 4. Streamlines are included again to show the relationship between the rotational structure and the velocity field.

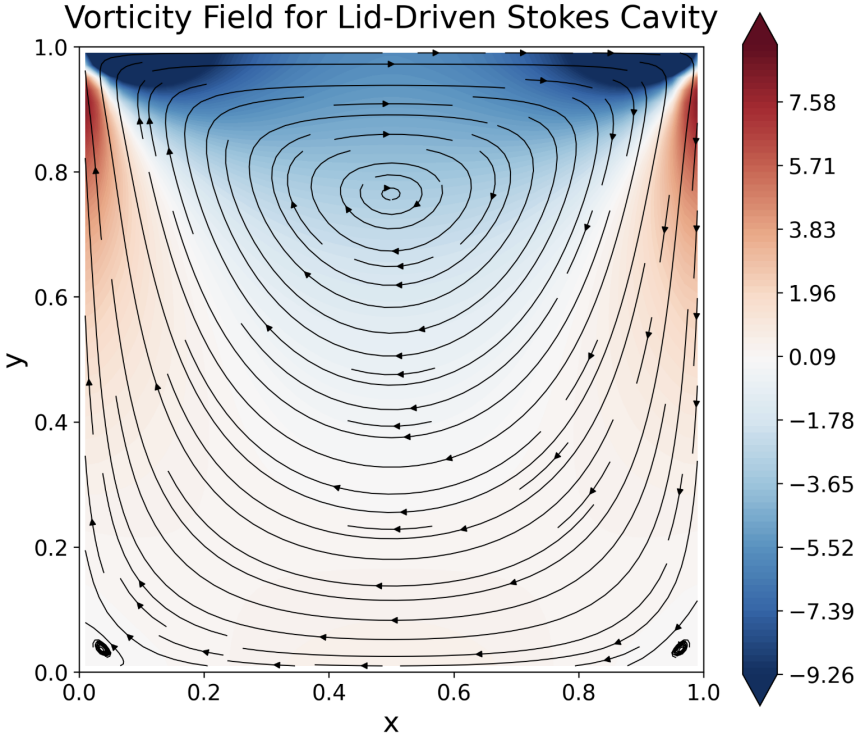


Figure 4: Vorticity field and streamlines for the lid-driven Stokes cavity.

The vorticity plot highlights behavior that is less apparent from the velocity magnitude alone. Strong vorticity develops near the moving upper wall because of the large velocity gradient between the moving lid and the interior fluid. Strong variations also appear near the upper corners, where the moving lid meets the stationary side walls. The interior streamline pattern corresponds to the large clockwise recirculation observed in Figure 3.

Conclusion

This experiment demonstrates that the staggered-grid Stokes solver is not limited to the manufactured periodic problem used for verification in Problems 1 and 2. By modifying the boundary treatment while retaining the same underlying MAC-grid formulation, the solver can be applied to a qualitatively different flow problem. The lid-driven cavity produces the expected recirculating flow, and computing the vorticity provides an additional way to investigate the structure of the numerical solution.